Functions:

- Thiamine act as a coenzyme in the form of thiamine pyrophosphate (TPP) in many enzymatic reactions. These are involved mainly in the breakdown of glucose to yield energy.
- Thiamine pyrophosphate (TPP) also acts as a coenzyme for transketolase reactions in the phosphogluconate oxidative pathway of carbohydrate metabolism. This reaction is essential for ribose formation, a constituent of DNA and RNA.
- Vitamin B_1 is also required in tryptophan metabolism for the activity of the enzyme tryptophan pyrolase.
- The adequate level of thiamine provides healthy nerves, a good mental outlook, a normal appetite and digestion.

Absorption and storage:

Free thiamine is readily absorbed by the small intestine. Thiamine is not stored in the human body. The excess thiamine is excreted in urine and also degraded by the enzyme thiaminase.

Deficiency:

The symptoms of thiamine deficiency occur because the tissue cells are unable to receive sufficient energy from glucose. Therefore, they cannot carry out their normal functions. Early symptoms of thiamine deficiency include fatigue, irritability, depression and numbness of the leg and poor tone of the gastrointestinal tract together with constipation.

Beriberi is the disease caused by thiamine deficiency. Beriberi is characterized by oedema in the legs. Usually, beriberi is caused by carbohydrate rich and low thiamine diets such as polished rice.



Figure 8.14 Deficiency due to vitamin B1

ii. Vitamin B_2 (Riboflavin)

Vitamin B_2 which is also known as riboflavin consists of a heterocyclic isoalloxazine ring attached to ribitol (a pentose alcohol). It is a yellow coloured compound.

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